

Demonstrating How to Increase the Dexterity of Electrical Muscle Stimulation using Back of the Hand Actuation

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Figure 1: We demonstrate how to improve the dexterity of the finger flexion produced by interactive devices based on electrical muscle stimulation (EMS). The key to achieve it is that we discovered a new electrode layout in the *back of the hand*. Instead of the existing EMS electrode placement, which flexes the fingers via the flexor muscles in the forearm, we stimulate the *interossei/lumbricals* muscles in the palm. Our technique allows EMS to achieve greater dexterity around the metacarpophalangeal joints (MCP), which we demonstrate in a series of applications, such as (c) a two-stroke drum roll or (d) a barred guitar fret. These examples, as with all others in our paper, were previously impossible with existing EMS electrode layouts.

ABSTRACT

We demonstrate a technique that allows an unprecedented level of dexterity in electrical muscle stimulation (EMS), i.e., it allows interactive EMS-based devices to flex the user's fingers *independently* of each other. EMS is a promising technique for force feedback because of its small form factor when compared to mechanical actuators. However, the current EMS approach to flexing the user's fingers (i.e., attaching electrodes to the base of the forearm, where finger muscles anchor) cannot flex one finger alone at the metacarpophalangeal (MCP) joint, they always induce *unwanted* actuation to adjacent fingers. To tackle the lack of dexterity, we propose a new electrode layout that places the electrodes on the *back of the hand*, where they stimulate the interossei/lumbricals muscles in the palm, which have never received attention with regards to EMS. We demonstrate the improved dexterity with a series of EMS-assisted music applications for playing piano, drums, and guitar.

CCS CONCEPTS

• **CCS CONCEPTS**; • **Human-centered computing** → Human computer interaction (HCI); Interaction devices; Haptic devices;

KEYWORDS

Electrical muscle stimulation, EMS, dexterity, haptic actuation

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1 INTRODUCTION

EMS is a promising technique for force-feedback & haptic-actuation because of its small form factor when compared to mechanical actuators (e.g., exoskeletons [1] and so forth). Therefore, EMS has been recently used to enable a wide range of haptic applications, such as: moving a user's wrist to tap to a beat [2, 3], teaching users how to manipulate objects they have never seen before [4], communicating information eyes-free [5] or even controlling the user's hand, which is holding a pen, to act as a plotter device [6].

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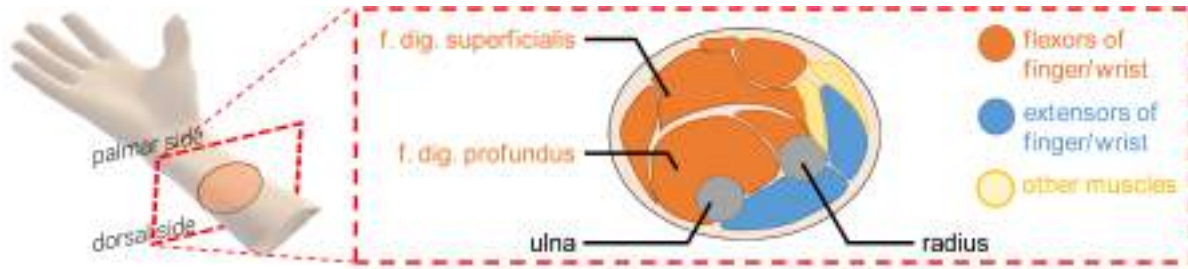


Figure 2: A model of cross-section muscle arrangement in the forearm and the hand, drawn from anatomy literature [9].

However, all these aforementioned EMS systems, and all others in the literature, are limited in that they either do not actuate the user's fingers (mostly only the wrist) or actuate the user's fingers very *coarsely* and not *independently* of each other, i.e., in most of these systems a particular target finger is flexed through EMS and other fingers start flexing as well, causing unwanted actuation and lack of *dexterity*—we denote this as the dexterity of the movements/poses induced by EMS on the user, not the user's own dexterity. The lack of dexterity in EMS-induced finger flexion was well documented in the *PossessedHand* [7], a work that pioneered the use of EMS to build interactive devices. In this system, the authors found that they could not flex any finger around the metacarpophalangeal (MCP) joint independently of the other fingers. Although 10 years have passed since the *PossessedHand*, it still stands as the most dexterous finger flexions in all interactive EMS devices.

We identify the cause of EMS's lack of dexterity when inducing a finger flexion, pinning it down to the fact that virtually every researcher and practitioner realizes it by placing electrodes at the forearm (standard-EMS). While this location intuitively makes sense, since all the finger flexor muscles are anchored there, it also is the cause for the lack of dexterity for three key reasons: (1) the flexor muscles in the forearm are *densely packed*, as they all meet at the elbow as a shared anchor; as such, attempting to flex a finger using electrodes attached to the skin causes currents to run also through an adjacent muscle, making other muscles move unwantedly; (2) the flexor muscles at the forearm are *layered* with finger flexors at one depth and wrist flexors at another; as such, attempting to flex a finger using electrodes attached to the skin often causes currents to run through the other layers as well, causing other muscles (e.g., wrist) to move unwantedly; (3) when we turn our wrist, the skin at the forearm *rotates by a different amount* compared to the muscles at the forearm; thus, whenever the wrist moves, the electrodes end up stimulating a new muscle, causing unwanted flexions.

To address the lack of dexterity in EMS, we discovered a novel electrode arrangement, depicted in Figure 1 (a), that robustly enables individual finger flexion. Our approach differs from the standard-EMS electrode arrangement in that we place the EMS electrodes on the *back of the hand* and instead stimulate the interossei/lumbricals muscles in the palm.

Surprisingly, we found through a user study [8] that this new electrode arrangement provides four key benefits when compared to the standard-EMS arrangement: our technique (1) flexes all four fingers around the metacarpophalangeal joint (MCP) more independently; (2) has less unwanted flexion of other joints (such as

the proximal interphalangeal joint); (3) is more robust to wrist rotations; and (4) reduces calibration time.

2 PRIOR WORK: THE “STANDARD-EMS” METHOD TO STIMULATE THE FINGER FLEXORS

Figure 2 depicts the cross-section of the forearm. The key insight that drives the standard-EMS arrangement is that the finger flexor (MCP) muscles are *packed densely* adjacent to one another and also *layered* on top of each other inside the forearm. Therefore, researchers have been using this location in virtually any EMS system that achieves finger flexion.

To illustrate the wide-spread use of this electrode placement, we found that in just the last decades of EMS research in HCI/Haptics, 54 publications have used, tested, described, or proposed applications that use this exact electrode placement on the forearm to achieve finger flexions (see [8] for details). Following, this arrangement emerged as a de-facto “standard-EMS” placement. In fact, like all our community, we have used this arrangement extensively in our own EMS research in both HCI and neuroscience.

3 OUR APPROACH FOR PRECISE FLEXOR ACTUATION: EMS ON THE BACK OF THE HAND

We solve the lack of dexterity in EMS by discovering a new electrode arrangement. Instead of placing the electrodes at the base of the arm to target the finger flexors at that location, we stimulate the back of the hand, targeting the unexplored lumbricals/interossei. As we found in our empirical study, this results in more dexterity when flexing fingers around the MCP joint using EMS.

The key insight that enables our approach is depicted in Figure 3. Interestingly, the hand's dorsal side has no extensor muscles and most of the dorsal muscles are interossei and lumbricals muscles, which happen to be MCP *finger flexors*. Importantly, these muscles are *close to the surface* and *spaced from each other*, making them ideal targets for EMS activation by placing electrodes on the back of the hand. While this might appear as trivial in hindsight, it has been completely overlooked and presents an immediate and significant impact on the dexterity of any EMS-based interactive devices that use any type of finger flexions.

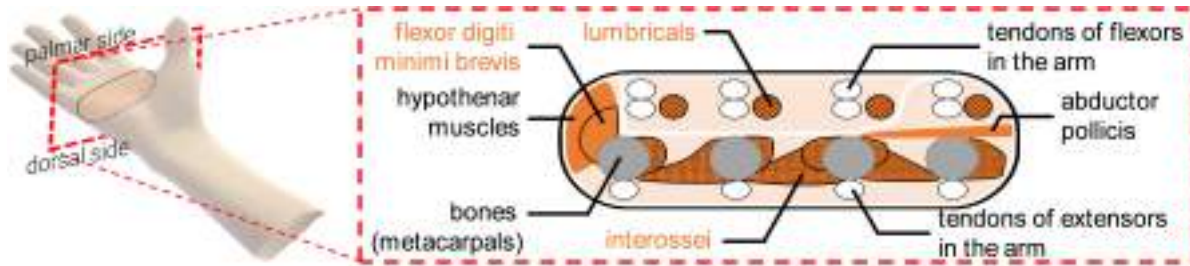


Figure 3: The key insight that enables our approach to achieve dexterity is that the lumbrical and interossei muscles, at the back of the user's hand, are more easily accessible to EMS than the standard forearm flexor muscles.

4 DEMONSTRATIONS OF DEXTROUS HAPTIC APPLICATIONS ENABLED BY OUR TECHNIQUE

Now that we validated that our back-of-hand EMS approach affords more dexterity by allowing us to flex fingers more independently [8], we explore some of the applications that our contribution enables. Note that all these applications are novel in that they are simply not possible with the standard EMS approach because all of these require individual finger actuation with significant precision. For instance, while researchers have used EMS for drumming by coarsely moving the user's wrist or biceps up/down, we demonstrate how to drum using a professional drumming technique, the double stroke roll, which requires individual control of the ring finger.

4.1 Actuating double-stroke drum rolls with EMS

After drummers spend some time acquainting themselves with the basic stroke (hitting the drum pad with one swift motion of their wrist) they start exploring how to drum faster. This is typically achieved with a technique called double-stroke drum roll (Figure 4). Unlike the basic stroke, double-stroke rolls require a high level of dexterity, in which the motion of the second stroke is entirely driven by a very precise movement of the MCP joint alone of the ring finger; this is used by drummers using the so-called "French grip" to grab the drumstick. We implemented a simple haptic application, depicted in Figure 4, which actuates users in performing the two-stroke roll.

4.2 Actuating fingers independently via EMS to play piano notes and chords

Much like all applications in our paper, this is a long-sought application in EMS that was never realized due to the lack of dexterity. However, we can now realize it via the back of the hand stimulation. In Figure 5, we depict a user who is being actuated to play (a) a piano melody with EMS finger flexions around the MCP joint for each finger and plays (b) a perfect fifth interval by having the EMS flex their fingers to correctly play two notes at the same time. While our technique provides the first step towards EMS for piano playing by flexing the user's fingers around the MCP joint, it

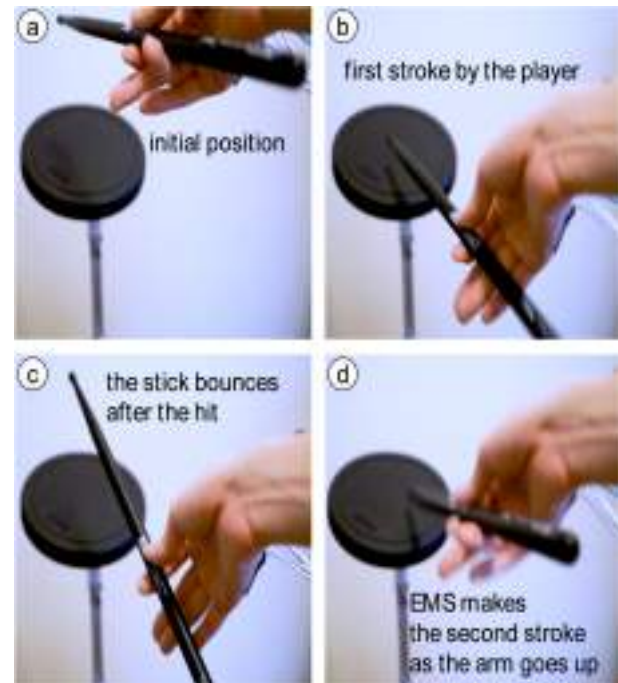


Figure 4: This drummer is being actuated using EMS to perform a double-stroke roll, a movement that requires significant finger independence (a quick finger flexion of the ring finger around the MCP joint) to realize.

does not actuate all finger joints required to play piano in its full expression.

4.3 Actuating fingers independently to play barred frets on the guitar

For playing the guitar, an important skill is to press all strings at a fret with the same finger (usually the index finger), which is required to make a *barre chord*—we chose these types of chords as they require only MCP flexion, unlike generic chords that typically require flexion of other joints. Figure 6 depicts an example of our simple haptic application that actuates users how to make barred frets by independently flexing their fingers around the MCP joint.

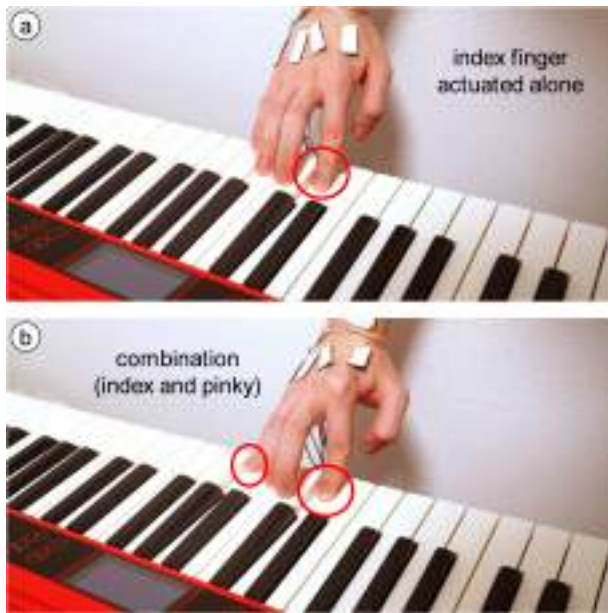


Figure 5: (a) With this newly achieved EMS dexterity, we can actuate a user to play simple piano melodies by having their fingers flexed, independently, using EMS. Also (b) we can assist a user in playing multiple notes and chords, such as this perfect fifth interval played, through EMS flexion of their index and pinky fingers at the MCP joint.

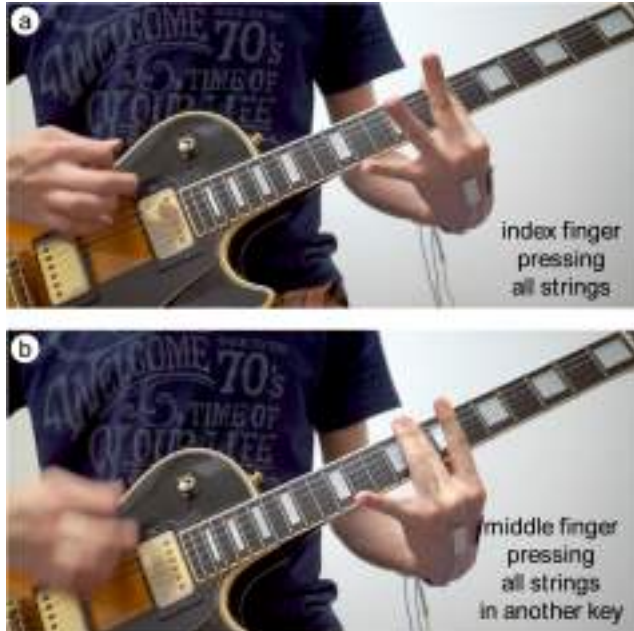


Figure 6: (a) Back-of-hand stimulation assists this user in performing a barred fret by flexing (a) their index finger and (b) their middle finger so that all strings at a fret are be pressed down completely.

5 CONCLUSIONS

We demonstrate a novel electrode layout that allows an unprecedented level of dexterity in electrical muscle stimulation (EMS), allowing interactive EMS-based devices to flex the user's fingers *independently* of each other. Our electrode layout places the electrodes on the back of the hand, where they stimulate the interossei/lumbricals muscles in the palm—a set of muscles that have been unexplored in EMS and that offer the following key improvements to the current EMS practice: (1) because these muscles are not as densely packed as the forearm flexors, our technique flexes all four fingers at the MCP joint more independently; (2) because these muscles are less densely layered than the forearm muscles, there is less unwanted flexion of other joints (such as from the proximal interphalangeal joint); (3) because these muscles are not subjected to forearm rotations, our technique is more robust to wrist rotations; and (4) because of all of the above, our technique reduced calibration time. We validated all these improvements through a user study [8].

To sum up, our EMS technique enables applications for interactive EMS systems that required a level of flexion dexterity *unachievable until now*. We demonstrate its dexterity with a series of musical applications for playing piano, drums, and guitar.

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